

# Christophe de Dinechin

Software as a Passion

SYSTEMS | LANGUAGES | TOOLS | GAMES | RESEARCH | COMMUNICATION SKILLS

## ◈ ABOUT

Pushing the boundaries by writing highly innovative and differentiating software, hoping to change the world one step at a time.

French software developer — operating systems, languages, games

Amateur physics researcher and tinkerer in strange forms of art

## ◎ KEY SKILLS

- ✓ Systems & Virtualization
- ✓ Languages & Compilers
- ✓ Tools & Apps
- ✓ Games & 3D
- ✓ Research
- ✓ Communication Skills

## ▤ TECHNOLOGIES

C | C++ | Virtualization | Kernel | Real-time | 3D | AI | SoC | Validation | Performance | Linux | macOS | OpenShift | RPL | XL | Tao3D | ELFE | SPICE | Make

## i INFO


★ MS Computer Science, École des Mines (1991, Paris)

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⌘ github.com/c3d

◉ c3d.github.io


**Scan the constellation**
https://c3d.github.io/cv/

## SYSTEMS & VIRTUALIZATION

**HPVM - Enterprise-grade Itanium virtualization** 2000-2010
- From first line of code and pre-silicon bring up all the way to 7 digits of revenue per quarter

**HP ECUTEST Real Time Test System for car electronics** 1994-1998
- Engine control unit (ECU) real-time testing and validation, 400 analog and digital I/O signals

**SPICE Remote Streaming** 2017-Present
- Bring the SPICE remote-access application to the modern 3D and video desktop

**DxO ONE Connected Camera** 2015-2016
- A tiny, but high-quality camera for photographers. Dual-core ARM + ISP, RTOS + Linux

## LANGUAGES & COMPILERS

**HP aC++ Compiler** 1998-2000
- First native pre-silicon Itanium compiler, exception handling, templates, performance, build system

**C++ ABI** 1999-2000
- Initiated a cross-industry standardization effort for C++ compiler interoperability, still in use today

**XL** 1992-Present
- XL is an Extensible Language with interpreted and JIT modes

**XLR8** 2026-Present
- XLR8 is a super-fast VM with interpreted and JIT modes

**DB48X - Modern RPL Calculator** 2022-Present
- Best scientific calculator today, a modern rebuild of high-end HP-48 and HP-50 calculators

## TOOLS & APPS

**Make-It-Quick** 2018-Present
- A make-based build system with auto-configuration, replaces automake/autoconf/cmake

**Recorder** 2017-Present
- A flight recorder for C and C++ programs, printf syntax, continuous logging, real-time graphing

**Emacs on Aqua** 2001-2005
- First graphical port of Emacs on macOS X (Rhapsody), first Emacs with transparent windows

## GAMES & 3D

**Alpha Waves / Continuum** 1989-1990
- First 3D platform game ever (Guinness Book of Records), precursor of Alone in the Dark

**HP-48 Games** 1989-1993
- Pacman, Tetris, Lemmings, demos for a cross-compiler I had written, went viral over infrared

**Tao3D** 2010-Present
- Tao3D is an XL-based platform for interactive real-time 3D presentations

**Taodyne** 2010-2016
- A company selling Tao3D and high-end interactive 3D presentations, stereoscopy, glasses-free 3D

## RESEARCH

**Computer science research**
- Published CS research: WIESS 2000 C++ exception handling; FOSDEM 2024 / FOSDEM 2023 talks on DB48X

**12 patents awarded, 3 pending**
- Covering virtualization, enterprise computing, confidential computing

**Physics research**
- A Theory of Incomplete Measurements — unifying GR and QM (TIM, Réunifions la physique)